

Diego Pilares Gallego

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ABOUT ME

Industrial engineer with an MSc in Motorsport Engineering from Oxford Brookes University, combining strong foundations in mechanical design, manufacturing processes, and data-driven optimisation. Experienced in CAD modelling, FEA, and dynamic system analysis, with a hands-on approach to solving complex engineering challenges. I'm passionate about innovation, quality, and continuous improvement, I thrive in collaborative, fast-paced environments focused on delivering efficient and reliable products.

EDUCATION

MSc Motorsport Engineering

Sept. 2024 – Sept. 2025; Oxford, UK

Oxford Brookes University

- Expected mark: Distinction (Dissertation submitted, awaiting final grade)
- Key modules: Advanced Vehicle Dynamics – Laptime Simulation and Race Engineering – Advanced Vehicle Aerodynamics – Composite Design and Impact Modelling.

BEng Industrial Engineering

Sept. 2019 – Jan. 2025; Madrid, Spain

Universidad Francisco de Vitoria – Motor & Sport Institute

- Final Mark: 6.91/10
- Key modules: Vehicle Dynamics – Suspension – Aerodynamics – Transmission and Gearbox – Powertrain
Fluid Dynamics – Thermodynamics – Machines and Mechanisms Theory – Design, Prototyping and Testing

PROFESSIONAL EXPERIENCE

Formula Student Team Member

Sept. 2024 – May 2025; Oxford, UK

Oxford Brookes Racing

- Collaborating with the Suspension & Structures departments to design and test new brake rotors and hub components.
- Using SolidWorks for CAD and HyperMesh for FEA, to enable structural and thermal analysis, as well as topology optimization.

Industrial Engineer Placement – Automotive Department

Feb. 2024 – Sept. 2024; Madrid, Spain

NTT DATA

- Involved in the development of corrective measures and new functionalities. Produced technical documentation, training presentations, analysed and resolved issues for the portals of BMW Spain, Portugal, Italy, Mexico, Brazil and India. Created layouts for Hyundai Spain.

Formula Student Team Member

Sept. 2019 – July 2021; Madrid, Spain

UFV Racing

- Collaborated with the Electronics department, creating wiring diagrams and PCB layouts via LTSpice and EasyEDA, developed Aim PDM integration and engine mapping.
- Collaborated with the Systems and Testing department to install powertrain, steering, suspension and brake components onto the vehicles, CAD of components in SolidWorks, 3D printing of test components via Ultimaker Cura and TIG welding.

MOST RELEVANT PROJECTS

MSc Dissertation: Hydrogen-fuelled LTC/CAI-IC engine control strategy optimization

Oxford Brookes University

Jan. 2025 – Sept. 2025

- 1D LTC simulation and ML-based control optimisation using GT-Suite to maximise hydrogen efficiency and reduce NOx emissions.

Next-gen F1 compliance and setup study

Oxford Brookes University

Apr. 2025 – May. 2025

- 2026-spec F1 vehicle setup validation using AVL VSM and AVL Drive for suspension, tyre and brake balance optimisation.

LMGT3 DrivAer front aero devices design and CFD validation

Oxford Brookes University

Oct. 2024 – Dec. 2024

- Designed and validated front aero components in SolidWorks and Star CCM+, improving aerodynamic performance.

Suspension Dynamics and Ride Optimisation

Oxford Brookes University

Oct. 2024 – Dec. 2024

- Developed bicycle, twin-track and 1, 2 and 4 DoF models in MSC Adams, to analyse and optimise cornering and vertical load response.

SKILLS

- **Data acquisition and analysis:** AVL Drive Race, MoTeC i2.
- **Vehicle and systems simulation:** AVL VSM, GT-Suite, MSC Adams, MATLAB, Simulink.
- **CAD, FEA and CFD:** SolidWorks, Star CCM+, XFLR5, Femap, Hypermesh.
- **Productivity:** Microsoft Office, Google Workspace.
- **Soft skills:** Analytical thinking, teamwork, adaptability, project management, attention to detail, critical problem solving, and technical communication.

LANGUAGES

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| • English - Fluent | • Spanish – Fluent |
| <i>Engineering MSc at Oxford Brookes</i> – 2025 | <i>Native</i> |
| <i>Pearson PTE Academic Overall Score 80</i> – 2024 | |
| <i>Cambridge English Certificate level B2</i> – 2019 | |

INTERESTS

Motorsport Engineering: Deep interest in vehicle design, performance optimisation and race technology evolution.

Sim Racing & Go-Karting: Exploring setup theory, tyre behaviour and handling dynamics.

Retro Electronics Restoration: Repair and modification of video game consoles and other electronics through reverse engineering and creative problem solving.

Gym training: Committed to training routines to support focus, resilience and discipline.

REFERENCES AND EMPLOYMENT STATUS

EU resident with a UK Student Visa until January 2026, references available upon request.